Donna George

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Technical Skills

- **Programming** Python, Java, C, C++, SQL
- **Machine Learning** Supervised Learning, Model Selection, Feature Engineering, Hyperparameter Tuning, Experimentation
- **Statistics & Analysis** Statistical Reasoning, Exploratory Data Analysis (EDA), Performance Metrics, Validation
- **Libraries & Tools** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Git, GitHub
- **Automation & Platforms** Automation Anywhere, Tkinter

Projects

- **Predicting Prices of Agri-Horticultural Commodities** Nov 2025
  - Defined a supervised machine learning approach to forecast agricultural commodity prices using historical market data.
  - Performed exploratory data analysis to identify trends, seasonality, feature relevance, and class imbalance.
  - Built an end-to-end pipeline using encoding, feature scaling, and Random Forest Regressor with an 80–20 train-test split.
  - Validated models using accuracy, ROC-AUC, and confusion matrix analysis, achieving 0.79 accuracy and 0.87 ROC-AUC.
  - Interpreted model outputs to assess prediction reliability and highlight limitations impacting real-world decision-making.

  Tech Stack: Python, Pandas, Scikit-learn, Matplotlib
- **AI-Driven Modulation Classification in Wireless Communication Systems** Jul 2025
  - Designed an ML-based modulation classification system for BPSK, QPSK, QAM, and FSK signals across varying SNR conditions.
  - Conducted experimentation using feature engineering, PCA, scaling, and GridSearchCV across multiple classifiers.
  - Improved classification accuracy to 93% using SVM and reduced low-SNR misclassification by 28% through iterative tuning.
  - Compared experimental results across models to select approaches balancing accuracy, robustness, and computational cost.

  Tech Stack: Python, Scikit-learn, NumPy, Pandas, Matplotlib
- **CPU Scheduler Simulator** Apr 2025
  - Implemented FCFS, SJF, Round Robin, and Priority Scheduling algorithms to simulate operating system process scheduling.
  - Built a Tkinter-based GUI with Gantt charts and automated computation of waiting, turnaround, and response times.
  - Analyzed scheduling trade-offs to compare algorithm performance under different workload conditions.

  Tech Stack: Python, Matplotlib, Tkinter

Training

- **Lovely Professional University** Jun 2025 – Jul 2025
  AI & ML for Real-World Problem Solving
  - **Applied Learning** Applied machine learning techniques to real-world datasets involving ambiguity and noisy data.
  - **Tech Stack** Worked with Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn for modeling and analysis.
  - **Optimization** Addressed class imbalance, feature overlap, and hyperparameter optimization challenges.

Certifications

- HackerRank Python (Basic) Nov 2025
- Certified Essentials Automation Professional – Automation Anywhere Nov 2025
- Basic Python Towards ML/AI – CSE Pathshala Mar 2024

Achievements

- **Dean's List (Top 10%) – Lovely Professional University** Nov 2025
  Distinguished for academic excellence by securing a position among the top 10% students university-wide based on overall academic performance.
- **Solved 150+ Leetcode problems** Feb 2026

Education

Bachelor of Technology in Computer Science Engineering Aug 2023 – Present
CGPA: 8.91

Higher Secondary Education (Class XII) 2019-2021
Percentage: 98%

Secondary Education (Class X) 2018-2019
Percentage: 98%